



IISRO

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Phoenix 🦅

Team Leader Name : Anushka Dabhade

Problem Statement : PS 12 - Fill in the Frames Seamlessly - Enhancing Temporal Resolution of Satellite Imagery using the techniques of AI/ML based on Optical Flow

Team Members

Team Leader: Name: Anushka Dabhade College: New LJ Institute of Engineering	Team Member-1: Name: Krithik Naidu College : New LJ Institute of Engineering
Team Member-2: Name: Deval Thakor College: New LJ Institute of Engineering	Team Member-3: Name: Fagun Soni College: New LJ Institute of Engineering

The Gap Nobody Else Closes

How our approach differs from every optical-flow and video-interpolation tool in use today.

WHAT'S OUT THERE TODAY

Traditional Optical Flow

Farneback / Lucas-Kanade break the moment a cloud rotates, forms, or dissolves.

Generic Video Interpolation

RIFE / Super SloMo are cinema tools. single-scale, no physical meaning preserved.

WHAT WE BUILT INSTEAD

1

Multi-Scale Optical Flow

Mesoscale drift + convective cells, modeled together

2

Cloud-Boundary Loss

Laplacian edge term keeps storm edges sharp

3

Per-Pixel Confidence Maps

Know exactly what to trust in every frame

4

GOES/Himawari → INSAT Transfer

Pretrain on 10-minute GOES/HIMAWARI satellite data, then fine-tune on INSAT data so the model can better adapt to regional weather conditions when applied.

5

Kelvin-Preserving Preprocessing

Physical meaning never sacrificed for convenience

6

Self-Supervised Training

Model learns temporal patterns directly from sequential satellite observations without manual labeling

7

Cloud Motion + Temporal Score

New metrics built for atmospheric motion

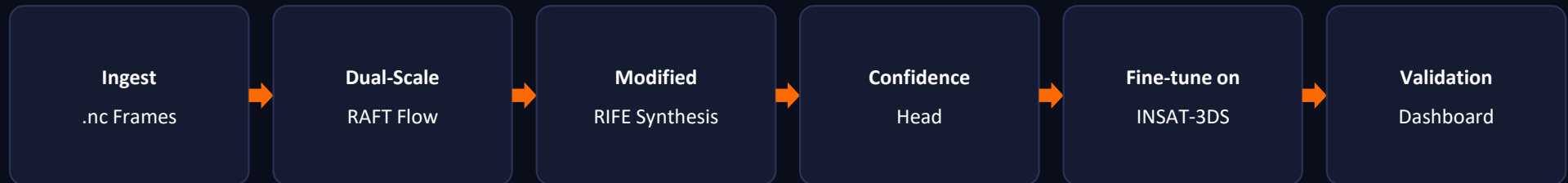
8

Generalizes Across Satellites

One pipeline : GOES, Himawari & INSAT alike

From Blind Spot to Trusted Forecast

INSAT-3DS photographs India every 30 minutes. A cyclone can jump Category 1 → 3 in 20 , invisible until now.



Built for Satellites, Not Cinema

Existing video interpolation approaches are not optimized for satellite-specific atmospheric dynamics. Ours is designed from scratch for thermal infrared physics.

Two Scales, Fused

Our dual-scale architecture jointly models mesoscale drift and convective-scale cloud motion for improved atmospheric motion estimation.

Confidence Maps That Operate

A frame without a reliability score is a demo. Per-pixel confidence turns this into something IMD could actually deploy.

"We don't interpolate pixels , we model atmospheric physics at multiple scales, and tell you exactly how much to trust each frame."

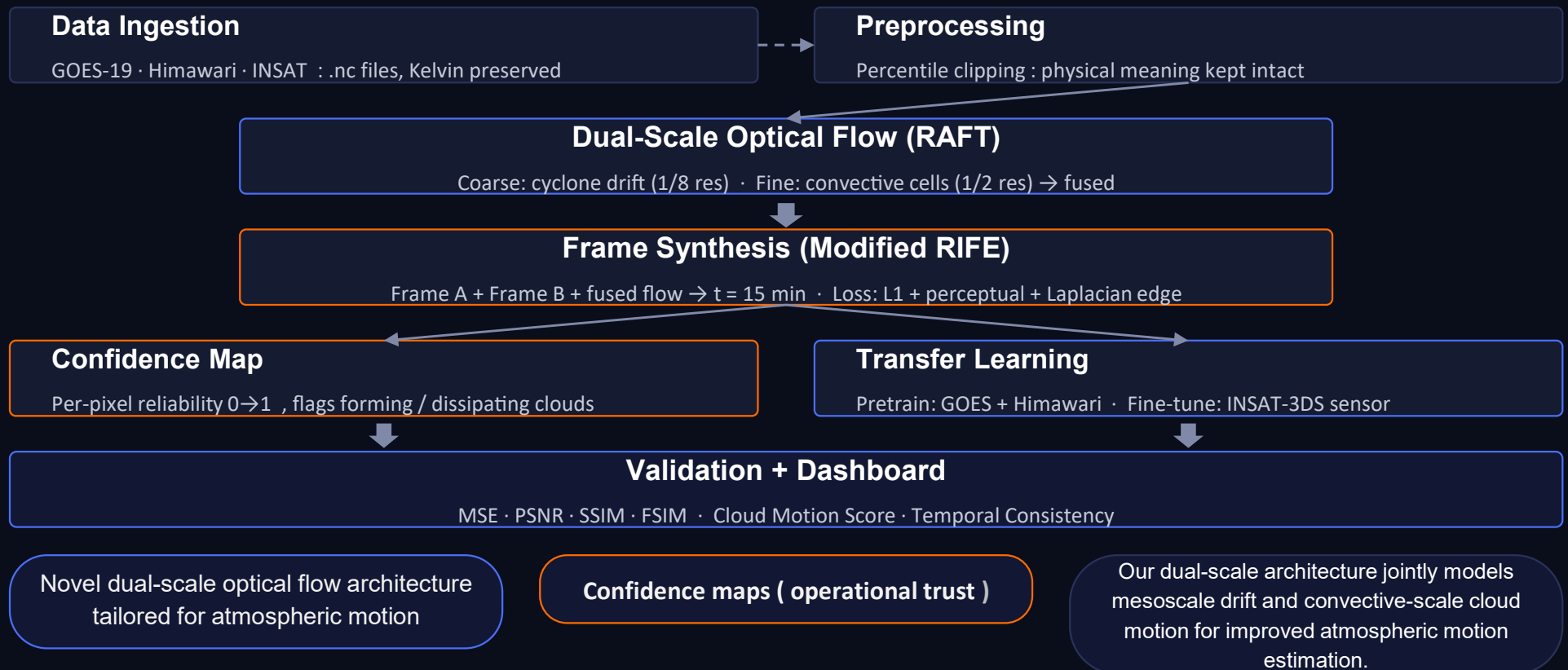
"Prototype will initially be trained on a smaller regional subset before scaling to full-disk satellite imagery."

What Our Solution Delivers (Features)



How It Works-End to End (Process flow Diagram)

From raw satellite data to trusted synthetic frames.



What the Meteorologist Sees(Mock Diagram/Wireframes)

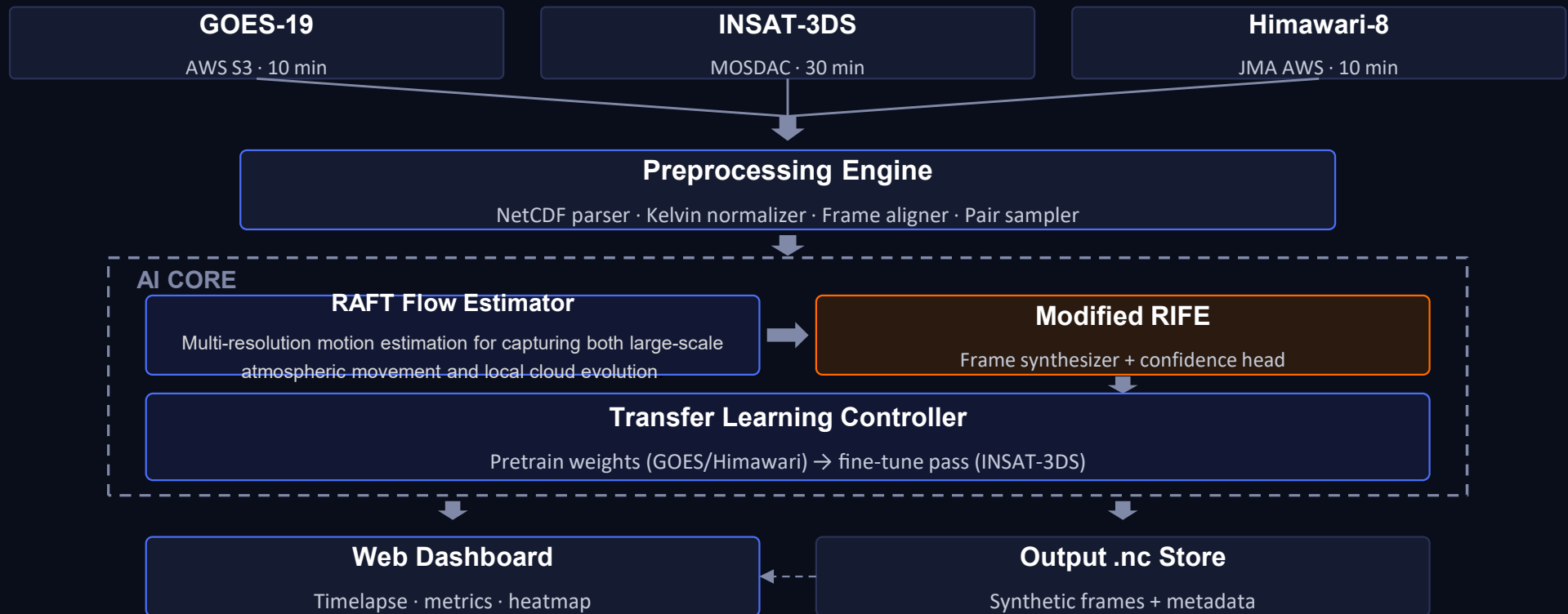
The dashboard : Real frames, synthetic frames, confidence, and metrics, all in one view.



"Every synthetic frame ships with a confidence map , so forecasters know exactly what to trust."

How the System Is Built (Architecture Diagram)

Four layers : Data, preprocessing, AI core, output.



Technologies Powering the Solution

Purpose-built stack — every tool chosen for a reason.

DATA LAYER

netCDF4

xarray

NumPy

Boto3

Sources: GOES-19 AWS S3 · MOSDAC INSAT-3DS · JMA Himawari-8

AI CORE

RAFT

RIFE (modified)

Confidence Head

Framework: PyTorch · Loss: L1 + perceptual + Laplacian edge

TRAINING

PyTorch Loop

W&B Tracking

Colab / Kaggle GPU

Free-tier compute, sufficient for initial fine-tuning

OUTPUT

React.js

Leaflet.js

Plotly

Flask API

Leaflet handles geospatial lat/lon overlays natively

EVALUATION

scikit-image

SSIM · PSNR · MSE

Cloud Motion Score

Temporal Consistency

Standard image metrics + two domain-specific scores

“No black boxes — every component is explainable, open-source, and reproducible.”



HACK2SKILL

BHARATIYA ANTARIKSH HACKATHON

2026

THANK YOU